Bridging the Language Gap: AI-Powered Identification of Yucatec Maya and Spanish for Emergency Response

Jean Carlos Buenfil Aguilar[1], Silvia Fernández-Sabido[2(✉)], Jorge Reyes-Magaña[3], Luis Basto-Díaz[3], Luis Fernando Curi-Quintal[3], and Oscar Gerardo Sánchez Siordia[2]

[1] Universidad Autónoma de Yucatán, Facultad de Matemáticas, Mérida 97000, Mexico
A21216396@alumnos.uady.mx

[2] Centro de Investigación en Ciencias de Información Geoespacial, Mérida 97302, Mexico
{sfernandez, osanchez}@centrogeo.edu.mx

[3] Universidad Autónoma de Yucatán, Facultad de Matemáticas, Mérida 97000, Mexico
{jorge.reyes,luis.basto,cquintal}@correo.uady.mx

Abstract. The wide linguistic diversity in Indigenous regions can create critical situations when these people need urgent medical or legal services and no one can identify the language they speak. Even if the institution has a directory of Indigenous interpreters, how do they know which one to contact? As a first step toward addressing this problem, this study explores linguistic identification between spoken Yucatec Maya and Spanish using deep learning models. A dataset of 3,820 audio files was selected, consisting of 2,116 clips in Spanish and 1,704 clips in Yucatec Maya. The audio files were segmented into 3 to 15 s fragments and gender-balanced where possible. RGB spectrograms were extracted from the audio segments for model input. Three deep learning architectures (ResNet-152, ResNet-50, and MobileNetV2) were evaluated to compare their accuracy, precision, recall, and F1 score performance. The results indicate that ResNet-152 achieved the highest accuracy (90.57%) and F1 score (0.9010), demonstrating its effectiveness in language identification tasks. MobileNetV2, although slightly less accurate (87.17%), showed superior efficiency, making it suitable for resource-limited environments. The learning curves indicate the need for more data or to expand the evaluation to other resource-limited languages to improve the generalizability of the results. The study highlights the potential of deep learning for resource-limited language identification and provides insight into the relationship between model performance and computational efficiency. These findings not only contribute to the development of tools for preserving Indigenous languages but can also expedite the search for human interpreters in emergency situations.

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1 Introduction

In Mexico, the National Institute of Indigenous Languages (INALI)[1] maintains a registry of interpreters and translators so they can be contacted in cases of health and legal emergencies. Given the linguistic richness of certain regions of the country, 68 languages and 364 variants [9], it is common for clinic and hospital staff, as well as courts and other legal bodies, to fail to correctly identify the language. There are real situations in which native people only communicate through their mother tongue and when they find themselves in medical or legal emergencies, it is difficult to communicate in official spaces with the corresponding professionals. Thus, the need arises to have a tool that allows the correct identification of the indigenous language of the vulnerable person. These situations delay the dispatch of interpreters to support critical situations where life and freedom are in danger. By leveraging advanced computational techniques, this study addresses the problem of language misidentification conducting initial tests to develop an effective and reliable system to distinguish between Yucatec Mayan and Spanish in spoken form.

2 Background

Indigenous languages, including the Mayan family (e.g., Yucatec Maya, Quiché, Tseltal), are spoken by millions of people in Mesoamerica, often in bilingual contexts alongside Spanish [24]. While this rich linguistic diversity represents a vital cultural heritage, it is common to encounter difficulties in accurately identifying and distinguishing these languages in oral communication.

Despite their cultural and social importance, Indigenous languages are underrepresented in technological applications, particularly in speech recognition and oral identification systems [17]. Existing tools focus primarily on the world's dominant languages, leaving a gap in support for Indigenous populations. This gap becomes especially problematic in high-stakes settings, such as healthcare, where medical personnel may have difficulty communicating with patients [6,13], or in legal proceedings that put people's freedom and integrity at risk [5]. In these contexts, it is essential to urgently contact appropriate interpreters, so distinguishing languages in the first few minutes of interaction is of utmost importance. This study seeks to contribute to addressing the lack of reliable oral identification systems through initial experiments with a classification model adapted to Yucatec Mayan and Spanish.

[1] https://www.inali.gob.mx.

3 Objective

In this phase, the objective is to evaluate the task of language identification using three deep learning models with distinct resource-performance trade-offs, applied to Yucatec Maya (a low-resource language) and Spanish.

4 Related Work

Indigenous language identification has become an increasingly important topic in natural language processing (NLP) as efforts to preserve and promote linguistic diversity grow. However, indigenous languages remain significantly underrepresented in computational research due to limited digital resources and a predominant focus on written languages. In [23], they highlight that most NLP research focuses on a small subset of Indigenous languages, leaving many without technological support. While progress has been made in machine translation and morphological analysis for selected Indigenous languages, significant gaps remain in key areas such as speech recognition, language identification, and resource development for low-resource languages. This section reviews existing research on Indigenous language identification and the use of deep learning models for audio and low-resource languages.

4.1 Language Identification

Language identification is a fundamental task in natural language processing and speech processing, with applications in multilingual systems, voice assistants, and low-resource language preservation. Traditional approaches rely on handcrafted features such as n-grams, phonotactics, and prosodic features [22]. However, these methods often struggle with low-resource languages due to limited annotated data and linguistic resources [11].

Recent advances in deep learning have enabled the end-to-end learning of language representations from raw audio or text, significantly improving performance in language identification tasks. A notable example is the deep learning-based approach, which uses convolutional neural networks (CNNs) for a language identification system powered by spectrograms. Experiments have been conducted for seven indigenous languages of India, yielding promising results in language classification [14].

4.2 Deep Learning for Audio and Low-Resource Languages

Deep learning models, particularly CNNs and recurrent neural networks (RNNs) have shown remarkable success in audio processing tasks, including speech recognition and language identification [1]. For low-resource languages, transfer learning and data augmentation techniques have been employed to mitigate the scarcity of labeled data [2]. Models like ResNet and MobileNet, originally

designed for image classification, have been adapted for audio tasks by treating spectrograms as visual inputs [25]. These models offer varying trade-offs between computational efficiency and accuracy, making them suitable for different deployment scenarios.

4.3 Gaps in the Literature

As noted by [21], most of the existing research focuses on written languages, whereas Indigenous languages are primarily oral. This focus on written language processing reveals a significant gap, as little to no research has been conducted on the recognition of Indigenous oral languages, despite their predominant spoken nature.

While deep learning models have been extensively studied for high-resource languages, their application to low-resource languages remains underexplored. Additionally, there is limited research comparing the resource-performance trade-offs of models like ResNet-152, ResNet-50, and MobileNetV2 in the context of language identification. This study aims to address these gaps by evaluating these models on the task of identifying Yucatec Mayan and Spanish, two languages with significant resource disparities.

5 Methodology

In this section, we describe all technical details applied to language identification. In the GitHub repo[2], you will find the source code of the elements described here.

5.1 Oral Corpus

The dataset comprises 3,820 audio files, including 1,704 in Yucatec Maya and 2,116 in Spanish. The Maya clips were obtained from the DoReCo [18], Maasab [10], and Taantsil [19] corpora while the Spanish audio clips were sourced from the Ciempiés [8] and Maasab corpora. In Table 1 we can see the description of each collection and the number of clips taken. The imbalance in the corpus can be solved by including more data in Maya to have better results. However, the present work is introductory, and digital resources for indigenous languages are limited. We want to highlight that Maasab contains recordings, in Mayan and Spanish, of doctor-patient dialogues included in two books by authors from Yucatán, Mexico [3,12].

5.2 Audio Feature Extraction

For audio manipulation, we used the Praat tool. Praat is a program that can be used to analyze, synthesize, and manipulate speech [4]. Initially, we explored the identification of vowel formants; however, this strategy requires making annotations on the vowels in the training audios. This is complicated for Mayan, which

[2] https://github.com/JeanBuenfil/Language-Classification-CNN.

Table 1. Corpus description

Collection name	Description	Maya clips	Spanish clips
Maasab	Doctor-patient phrases	785	798
DoReCo	Natural conversations	792	0
T'aantsil	Natural conversations	127	0
Ciempiés	Press, literature and conversations	0	1318

has 25 vowels that people pronounce differently depending on the region. For this reason, we opted to work with spectrograms.

Each audio clip was segmented into fragments ranging from 3 to 15 s to ensure consistency in length. Additionally, efforts were made to maintain a balanced representation of female and male speakers wherever possible. Following the selection and segmentation of the audios, RGB spectrograms were generated for further processing. Spectrograms were obtained in grayscale using the Librosa library with its default parameter settings. Each audio signal was loaded as a single-channel waveform with a sampling rate of 22,050 Hz. The resulting spectrograms were then converted to a decibel (dB) scale. A frequency resolution of 2048 and a hop length of 512 were used, covering a frequency range from 0 to 11,025 Hz. Finally, each spectrogram was converted to RGB format and resized to 224 × 224 pixels.

Figure 1 shows the sound signals and resulting RGB spectrograms for the Yucatec Maya and Spanish audios of a sentence from the Maasab corpus.

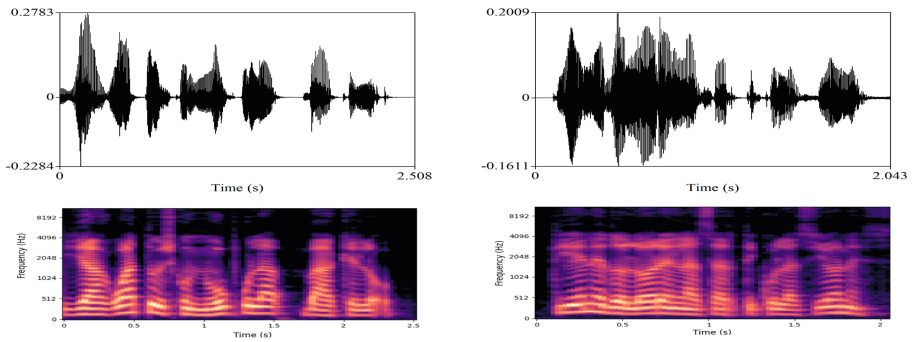


Fig. 1. Sound signal and spectogram (a) for Yucatec Maya sentence *Yaj wáaj tu'ux ku núupul a baakelo'ob*; sound signal and spectogram (b) for Spanish sentence *Â£ Tiene dolor en las articulaciones?*. Both mean "Do you have joint pain?".

5.3 Deep Learning Models

Since applications for low-resource languages are typically implemented on limited capacity devices, an experiment was conducted with three state-of-the-art models that have demonstrated good performance in image classification but that present different balances between accuracy and resource usage: Microsoftresnet-152 and Microsoftresnet-50 from the HuggingFace repository [20], as well as MobileNetV2 from the Deep-Learning repository [15]. Table 2 shows a comparison between them. The Microsoft/resnet models are convolutional neural networks with 50 and 152 layers, respectively, pre-trained on ImageNet-1k to classify images into 1,000 categories. It uses residual blocks to facilitate training deep networks [7]. MobileNetV2 is also a convolutional neural network, but specifically designed to reduce computational costs and improve performance on mobile devices. It uses an inverted residual structure, where residual connections are between bottleneck layers, and employs depth-light convolutions to filter features. It consists of an initial fully convolutional layer with 32 filters, followed by 19 residual bottleneck layers [16].

Table 2. Comparison of used models

Model	Parameters	Flop's	Use case	Key strength
Resnet-152	∼60 M	∼11.3 B	High-accuracy tasks	Depth/accuracy
Resnet-50	∼25 M	∼4.1 B	Balanced performance	Speed/accuracy trade-off
MobileNetV2	∼3.4 M	∼300 M	Mobile/edge deployment	Efficiency/speed

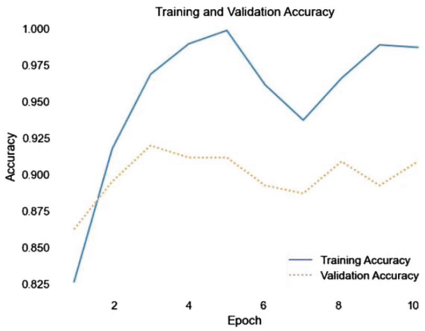
6 Results

In this study, three deep learning models (ResNet-152, ResNet-50, and MobileNetV2) were evaluated for the task of audio classification to identify two languages: Yucatec Mayan and Spanish. The corpus used consisted of 3820 audio samples, split into 80% training (3056 samples), 10% validation (382 samples), and 10% testing (382 samples), with a class imbalance (1704 Mayan audios and 2116 Spanish audios).

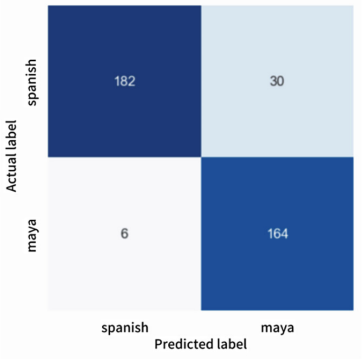
The results of accuracy, precision, recall, and F1-score are presented in Table 3. Figure 2 shows the training and validation accuracy across epochs and the confusion matrices. We consider the graphs as a useful visual tool for comparing the behavior of the three models employed.

7 Discussion

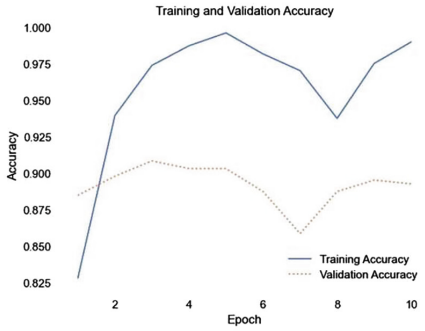
The results show significant differences in model performance, allowing for an analysis of their strengths and weaknesses in the context of language identification.



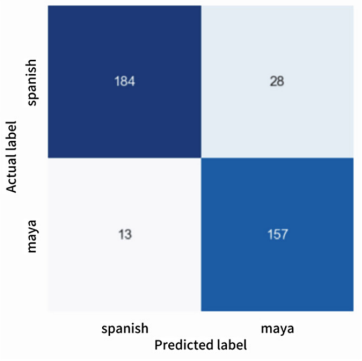
(a)



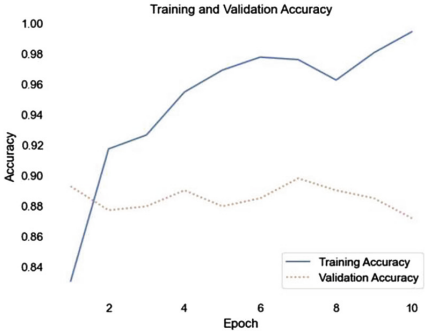
(b)



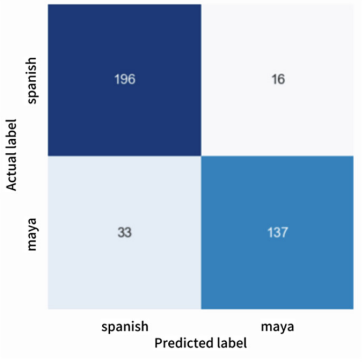
(c)



(d)



(e)



(f)

Fig. 2. Training-validation accuracy and confusion matrix for ResNet-152 (a, b), ResNet-50 (c, d) and MobileNetV2 (e, f).

Table 3. Language identification results.

Model	Accuracy	Precision	Recall	F1-score
Resnet-152	0.9057	0.8453	0.9647	0.9010
Resnet-50	0.8926	0.8486	0.9235	0.8845
MobileNetV2	0.8717	0.8954	0.8058	0.8482

7.1 Overall Performance

ResNet-152 achieved the best overall performance, with an accuracy of 90.57%, an F1-score of 0.9010, and a recall of 0.9647. This suggests that the model is highly effective at correctly identifying both languages, despite the class imbalance in the corpus. Its ability to generalize well across both classes (Maya and Spanish) is notable.

ResNet-50 showed slightly lower performance, with an accuracy of 89.26% and an F1-score of 0.8845. Its lower recall (0.9235) indicates that it struggles more to correctly identify all instances of the target languages, especially compared to ResNet-152. In the Fig. 2d we see an asymmetry in errors: 28 false negatives in Spanish vs only 13 false positives in Maya.

MobileNetV2 achieved an accuracy of 87.17%, slightly lower than ResNet-50, but with a notably lower recall (0.8058), suggesting that it is not very effective at capturing most instances of the target languages. However, its higher precision (0.8954) indicates a lower rate of false positives compared to the ResNet models as we can see in Fig. 2f.

The learning curves indicate that ResNet152 performs better than ResNet50, showing less overfitting despite its greater depth. ResNet50 and MobileNetV2 exhibit similar overfitting, but MobileNetV2 is lighter and more efficient. Although the overfitting of the models is in all cases $<10\%$ gap, it could be reduced by obtaining more data, which is sometimes difficult when dealing with resource-poor languages.

7.2 Corpus Imbalance

The corpus used in this study has a class imbalance, with 2116 Spanish audios and 1704 Mayan audios. This imbalance can affect model performance, particularly for the minority class (Mayan). However, the results show that the models achieve competitive performance for both classes: ResNet-152 handles the imbalance well, as evidenced by its high recall (0.9057) and F1-score (0.9010), indicating that it is effective at identifying both Spanish and Mayan. MobileNetV2, despite its lower recall, has the highest precision (0.8954), suggesting that it is particularly effective at identifying the correct class. This could be due to its lightweight architecture and ability to generalize better with limited data.

7.3 Resources Usage

Although ResNet-152 achieved the best performance in terms of accuracy and F1-score, its computational cost must be considered. ResNet-152 is a much larger and more complex model than ResNet-50 and MobileNetV2, requiring significantly more resources (memory, training time, and computational power). In contrast, MobileNetV2 is much more resource-efficient, making it better suited for environments with hardware limitations or real-time applications. We focused mainly on deep-learning approaches since some authors [14] performed experiments based on classical machine-learning techniques, like Naïve Bayes, SVM, Random Forest, etc. This shows that deep-learning approaches outperformed classical techniques.

7.4 Low-Resource Languages

In the context of low-resource languages like Yucatec Mayan, the choice of model depends on the specific application. If accuracy is the priority and sufficient computational resources are available, ResNet-152 is the best option. If a balance between performance and efficiency is desired, ResNet-50 is a strong alternative. For applications in resource-constrained environments or where inference speed is critical, MobileNetV2 is the most suitable option, despite its lower precision. Additionally, obtaining more data (or using data augmentation techniques) can help reduce overfitting and improve generalization across models.

Due to the limited availability of data for experiments involving low-resource languages, it is necessary to rely on multiple sources for corpus construction. As a result, incorporating features such as speaker gender identification or contextual information becomes challenging in the final corpus. Nonetheless, these features could prove valuable for identifying potential errors and enabling more meaningful comparisons in future, in-depth studies. When working with a broad range of speaker profiles, it is essential for the system to incorporate normalization techniques. Additionally, the use of diverse training data and speaker-invariant modeling strategies is crucial to maintaining both the effectiveness and inclusivity of the language identification system

8 Conclusion and Future Work

While the results are promising, some study limitations should be noted. Although the corpus of 3820 audios is sufficient for a preliminary study, a larger and more balanced dataset could improve model performance, especially for Yucatec Mayan. Further fine-tuning of hyperparameters could improve the performance of MobileNetV2 and ResNet-50, bringing them closer to the level of ResNet-152. Obtaining more data or extending the evaluation to other low-resource languages would allow for better generalization of the results.

In summary, the results demonstrate that ResNet-152 is the most effective model for the language identification task in terms of recall and F1-score. However, MobileNetV2 stands out for its efficiency and high precision, making it

ideal for applications in resource-constrained environments. The choice of model should be based on the desired balance between performance and efficiency, as well as the practical constraints of the deployment environment. Additionally, the class imbalance in the corpus did not significantly affect model performance, suggesting that the models are robust for language identification tasks with limited data.

Despite this work introduces preliminary results in language spoken classification between Yucatec Maya and Spanish, we want to state the importance it gives to under-represented Indigenous languages in México, a country with great linguistic diversity. The correct classification of language can bring multiple benefits to Indigenous communities who cannot express themselves in a language other than their mother tongue, so this technology can be of great interest in a system that first identifies the language used and then channels qualified translators to help people in vulnerable situations.

This work represents an initial step toward the automatic identification of spoken language. In future work, we aim to incorporate more advanced and contemporary approaches to spoken language identification. Additionally, to achieve broader impact and improved accuracy, it is essential to expand the corpus to include additional indigenous languages or their variants, always with full respect for the rights and consent of the individuals contributing their voices. Moreover, it is important to address ethical and social challenges, such as safeguarding the privacy of audio data and ensuring informed consent. Collaborating closely with Indigenous communities is also crucial to ensure that the system remains culturally sensitive and contextually appropriate. These considerations are vital not only for ethical integrity but also for building trust and promoting long-term sustainability in language technology initiatives. With the development of a robust and reliable model, future implementations could include integration into mobile applications for tablets and smartphones, thereby supporting accessibility and technological inclusion for underrepresented language communities.

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Systems Modelling and Simulation

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Editors: Nurul Adilla Mohd Subha, Norhaliza Abdul Wahab, Pei Ling Leow & Herman Wahid...

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Paper ID
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Paper Title
Bridging the Language Gap: AI-Powered Identification of Yucatec Maya and Spanish for Emergency Response

Track Name
COMIA2025

Reviewer #6

Questions

1. Realice un breve resumen del artículo
El presente manuscrito aborda un problema relevante y significativo: la identificación de idiomas en situaciones de emergencia, un tema crítico en contextos de diversidad lingüística como el de México. Emplea un enfoque basado en aprendizaje profundo para distinguir entre el maya yucateco y el español es innovador y tiene un claro potencial de impacto social.

2. Describa las fortalezas del artículo
Los resultados preliminares son sólidos, con una precisión destacable de alrrededor del 90% usando ResNet-152, y el uso de MobileNetV2 añade un ángulo práctico al considerar entornos con recursos limitados.

3. Describa las áreas de mejora del artículo
1. Claridad en la Introducción y Motivación
Observación: La introducción describe el problema de la identificación de idiomas en emergencias, pero no logra transmitir plenamente la motivación y urgencia del estudio. Aunque se menciona la diversidad lingüística y la existencia de un registro de intérpretes, no se profundiza en por qué este trabajo es crucial ni en sus implicaciones prácticas.
Sugerencia: Te recomiendo fortalecer la introducción con ejemplos concretos o datos que ilustren la magnitud del problema. Por ejemplo, ¿qué porcentaje de emergencias en regiones indígenas se ve afectado por barreras lingüísticas? Incluir una breve

referencia al impacto en la vida de las personas (como retrasos en atención médica o legal) y conectar el trabajo con la preservación de lenguas indígenas podría hacer la motivación más convincente y atractiva para los lectores.

2. Detalles Metodológicos

Observación: La sección de metodología ofrece una visión general adecuada, pero carece de detalles específicos que permitan la reproducibilidad. Por ejemplo, la extracción de características de audio y la generación de espectrogramas se mencionan brevemente, sin especificar parámetros clave utilizados. Además, aunque se reconoce el desequilibrio de clases en el conjunto de datos , no se describe cómo se abordó este problema en el entrenamiento de los modelos.

Sugerencia: Proporciona más detalles sobre el proceso de extracción de características, como los parámetros de configuración epara los espectrogramas RGB. Asimismo, explica si aplicaste alguna técnica para mitigar el desequilibrio de clases, como sobremuestreo, submuestreo o ajuste de pesos en la función de pérdida.

3. Análisis de Resultados

Observación: Los resultados son prometedores, con ResNet-152 alcanzando una precisión del 90.57%. Sin embargo, el análisis es limitado y no explora a fondo los errores de clasificación ni las posibles razones detrás de las diferencias en el rendimiento entre los modelos. Por ejemplo, las matrices de confusión muestran asimetrías en los errores, pero no se discuten en detalle.

Sugerencia: Realiza un análisis más profundo de los resultados. ¿Qué tipos de clips (por duración, género del hablante o contexto) fueron más propensos a errores? ¿Hay patrones en los falsos positivos o negativos que podrían guiar mejoras futuras? Además, una comparación de los modelos que incluya tiempos de entrenamiento y recursos computacionales podría enriquecer la discusión sobre sus compensaciones prácticas, más allá de las métricas de rendimiento.

4. Comentarios generales para los autores

Discusión sobre Implicaciones Prácticas

Observación: La discusión se centra principalmente en el rendimiento técnico de los modelos, pero dedica poco espacio a las implicaciones prácticas del sistema. No se aborda cómo podría integrarse en servicios de emergencia reales ni los pasos necesarios para su implementación en el mundo real.

Sugerencia: Amplía la discusión para explorar cómo este modelo podría usarse en la práctica. Por ejemplo, ¿se integraría en un sistema telefónico de emergencias o en dispositivos portátiles? ¿Qué infraestructura sería necesaria? También podrías considerar desafíos éticos o sociales, como la privacidad de los datos de audio o la colaboración con comunidades indígenas para asegurar que el sistema sea culturalmente sensible.

5. Cuál sería su evaluación para este artículo con base en su revisión

Aceptado

Reviewer #8

Questions

1. Realice un breve resumen del artículo

A proposal for binary recognition between Yucatec Maya and Spanish is presented, evaluating three convolutional network methods that consider spectrograms as input signals.

2. Describa las fortalezas del artículo

The incorporation of digital signal processing techniques.

3. Describa las áreas de mejora del artículo

In order to enhance the quality of the work, it is imperative to provide a robust theoretical and mathematical foundation. This approach ensures that the work does not remain superficial.

4. Comentarios generales para los autores

1. It has been recommended that the computational contribution be highlighted, as it is common practice to present the case study emphasizing the scientific component.

2. It is imperative to delineate the spectrogram size and its alignment with the stipulated network requirements.

3. The following inquiry is posed: How do spectrograms vary when a male, a female, a child, a male child, or an elderly adult speaks? In the event that the speakers are of different ages, what modifications must be made to ensure the functionality of the current proposal?

4. In the results section, the work is reduced to the recognition between Yucatec Mayan and Spanish. However, at the beginning of the document, it is mentioned that even in Yucatec Mayan there are variations conditioned to the regions. It is therefore necessary to extend the work in order to identify the various variants.

5. The findings depicted in Figure 2 are not suitable for presentation, as this is a prevalent issue in research papers that employ graphs and confusion matrices during the training phase. The inflation of these metrics can be attributed to overtraining. The results to be reported are those obtained in the test pass.

6. It is recommended that a minimum of five metrics be reported, thereby facilitating the establishment of more rigorous criteria for evaluating the quality of the result.

7. It is necessary to justify the rationale behind focusing exclusively on these convolutional architectures, while disregarding more contemporary ones.

5. Cuál sería su evaluación para este artículo con base en su revisión

Débilmente aceptado